

SECTION A

1 (**short**) A reversible heat engine takes in heat Q_H at temperature T_H and rejects heat to the atmosphere at temperature T_0 . It produces work, W , which drives a reversible heat pump, which takes in heat at temperature T_0 and rejects heat Q_R to a room at temperature T_R .

(a) Show that

$$\frac{Q_R}{Q_H} = \frac{1 - \tau_E}{1 - \tau_P}$$

where $\tau_E = T_0/T_H$ and $\tau_P = T_0/T_R$. [5]

(b) Sketch Q_R/Q_H as a function of τ_P for $\tau_E = 1/2$. [3]

(c) Use the result of (b) to state how the performance of domestic heat pumps can be maximized. [2]